Lab - 3: Experiment No: 3

Aim: Create a Cryptocurrency using Python and perform mining in the Blockchain created.

Objectives

- To understand the working of a cryptocurrency and blockchain.
- To implement a blockchain using Python.
- To create a decentralized Peer-to-Peer (P2P) blockchain network.
- To perform cryptocurrency transactions.
- To perform Proof-of-Work mining.
- To implement the longest-chain consensus mechanism.
- To remove pending transactions after they are added to a block.

Theory:

Blockchain

A blockchain is a distributed digital ledger that stores data in a sequence of blocks. Each block contains transaction information, a timestamp, a proof value, and the hash of the previous block.

The major components of a block in this experiment are:

- **Index** – Position of the block in the blockchain.
- **Timestamp** – Time at which the block was created.
- **Transactions** – Transactions included in the block.
- **Proof** – Number obtained through the Proof-of-Work process.
- **Previous Hash** – Hash of the previous block.

Each block is linked to the previous block using its hash. Therefore, changing the contents of a previous block would invalidate the subsequent blocks.

Cryptocurrency

A cryptocurrency is a digital currency that uses cryptographic techniques and blockchain technology to record transactions without requiring a central authority.

In this experiment, a simple cryptocurrency called **HadCoin** is implemented using Python.

A transaction contains:

Sender
Receiver

Amount

For example:

```
{
   "sender": "Khushi Shukla",
   "receiver": "Bob",
   "amount": 29
}
```

Transactions

When a transaction is submitted, it is not immediately stored inside a block. It is first stored in the node's pending transaction list:

self.transactions

When a miner mines a new block, these pending transactions are included in the block.

Mempool

A mempool is a temporary pool containing valid transactions that have been received by a blockchain node but have not yet been included in a block.

In this implementation:

self.transactions

acts as the pending transaction pool or simplified mempool.

After the transactions are included in a block, the list is cleared using:

self.transactions = []

Thus, the same transactions are not included repeatedly in subsequent blocks.

Mining

Mining is the process of creating a new block by solving a computational Proof-of-Work problem.

The program calculates:

hashlib.sha256(

```
    str(new_proof**2 - previous_proof**2).encode()
).hexdigest()
```

The mining process continues until the generated hash starts with:

0000

The proof that satisfies this condition is used to create the new block.

Mining Reward

The miner receives a reward for successfully mining a block.

In this implementation, the mining function adds:

```
blockchain.add_transaction(
    sender=node_address,
    receiver='Richard',
    amount=1
)
```

Therefore, a reward of **1 HadCoin** is included as a transaction in the mined block.

Multi-Node Blockchain

Three independent blockchain nodes were created:

Node 1 → Port 5001
Node 2 → Port 5002
Node 3 → Port 5003

Each node maintains its own copy of the blockchain and communicates with other nodes through HTTP requests.

Consensus and Longest Chain Rule

Different nodes may temporarily have different versions of the blockchain.

The replace_chain() function implements the consensus mechanism. It contacts the connected nodes, obtains their chains, checks their validity, and replaces the current chain if a longer valid chain is found.

Thus, the network eventually agrees on the longest valid blockchain.

Procedure / Steps Performed

Step 1: Download the Lab Files

The Lab 3 files were downloaded and extracted.

The following files were available:

hadcoin.py
hadcoin_node_5001.py
hadcoin_node_5002.py
hadcoin_node_5003.py
nodes.json
transaction.json

Step 2: Open the Lab 3 Directory

The Lab 3 folder was opened in PowerShell.
cd ".\Lab_3_Create a Cryptocurrency"

Step 3: Check Python Version

The Python version was checked using:
python --version
Python 3.14.7 was installed.

Step 4: Create a Virtual Environment

A virtual environment was created using:
python -m venv venv
Since PowerShell execution policy prevented direct activation, the Python executable inside the virtual environment was used directly.

Step 5: Install Required Libraries

Flask and Requests were installed using:
.\venv\Scripts\python.exe -m pip install Flask
and:
.\venv\Scripts\python.exe -m pip install requests
The installation was verified successfully.

Step 6: Run Node 5001

The first blockchain node was started using:
.\venv\Scripts\python.exe .\hadcoin_node_5001.py

The node started successfully at:
http://127.0.0.1:5001

Step 7: Run Node 5002

A second terminal was opened and Node 5002 was started:

.\venv\Scripts\python.exe .\hadcoin_node_5002.py

The node started at:

http://127.0.0.1:5002

Step 8: Run Node 5003

A third terminal was opened and Node 5003 was started:

.\venv\Scripts\python.exe .\hadcoin_node_5003.py

The node started at:

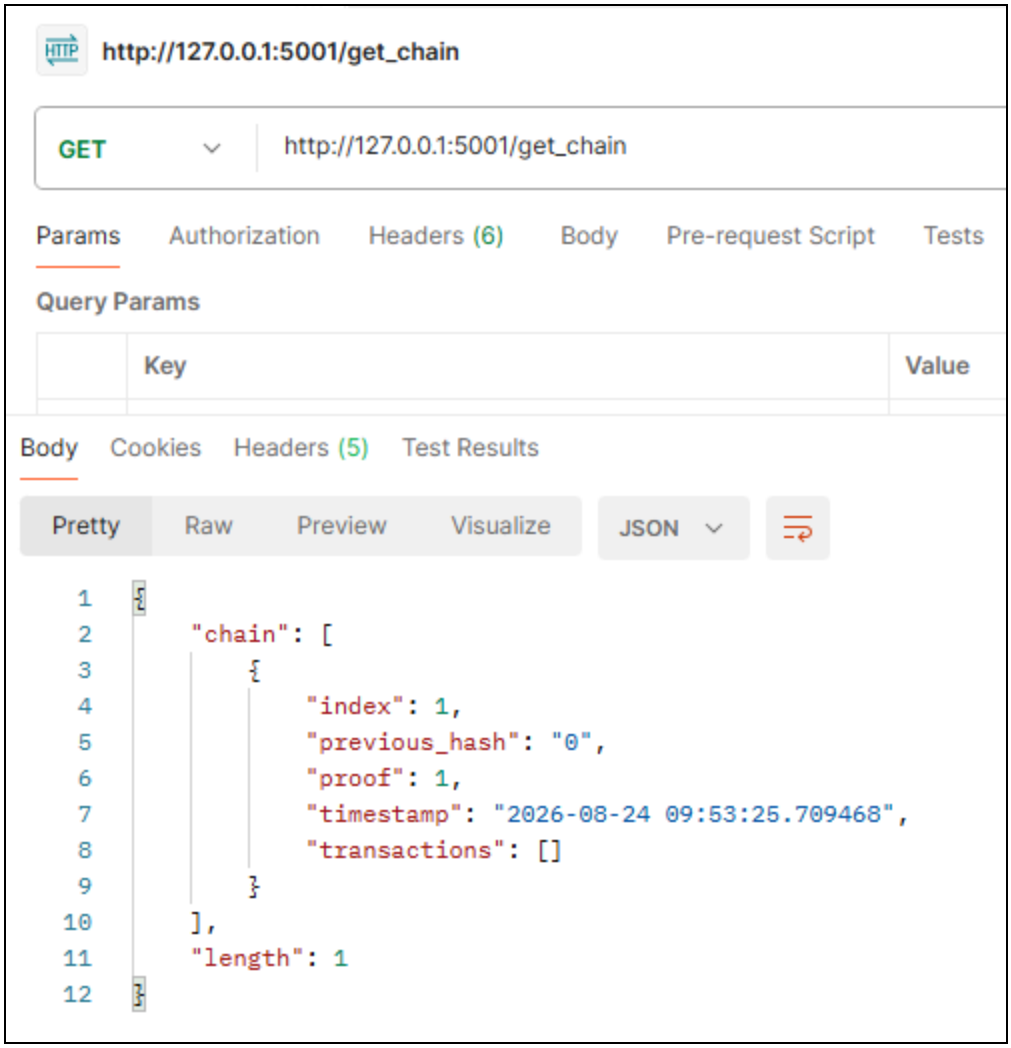
http://127.0.0.1:5003

Thus, three blockchain nodes were running simultaneously.

6. Testing the Blockchain Using Postman

Step 9: Fetch the Initial Blockchain

Khushi Shukla_29_D20A



Khushi Shukla_29_D20A

http://127.0.0.1:5002/get_chain

GET http://127.0.0.1:5002/get_chain

Params Authorization Headers (6) Body Pre-request Script Tests

Query Params

	Key	Value

Body Cookies Headers (5) Test Results

Pretty Raw Preview Visualize JSON


```
{
    "chain": [
        {
            "index": 1,
            "previous_hash": "0",
            "proof": 1,
            "timestamp": "2026-08-24 09:54:05.545717",
            "transactions": []
        }
    ],
    "length": 1
}
```


http://127.0.0.1:5003/get_chain

GET http://127.0.0.1:5003/get_chain

Params Authorization Headers (6) Body Pre-request Script Te

Query Params

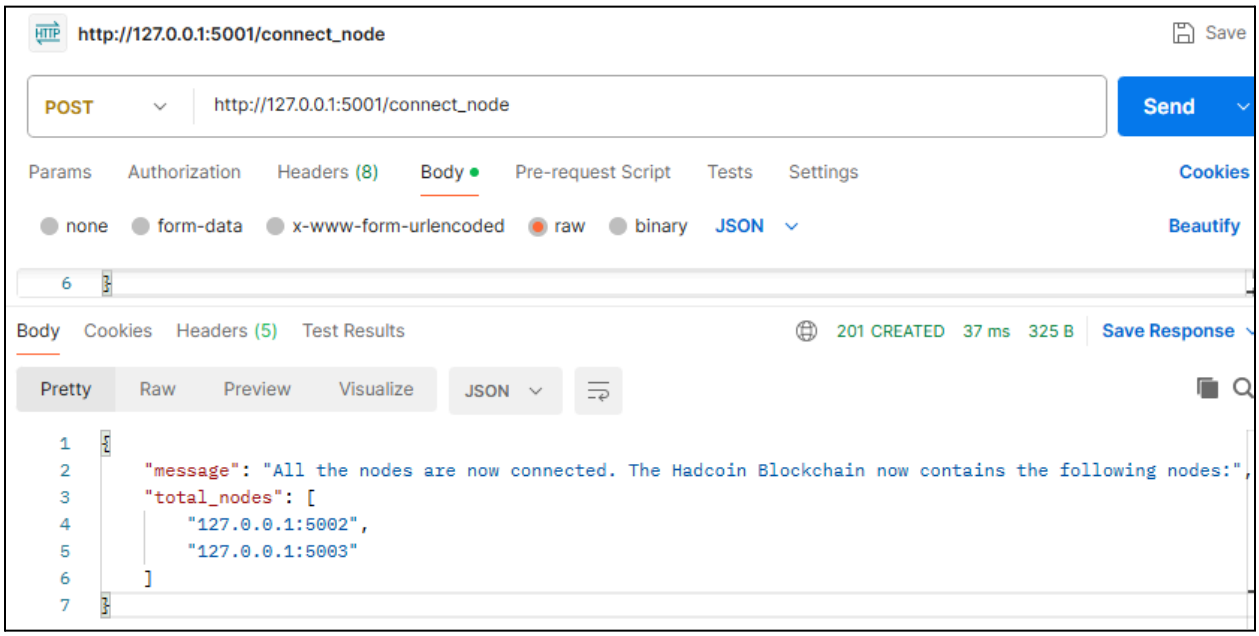
	Key	Val

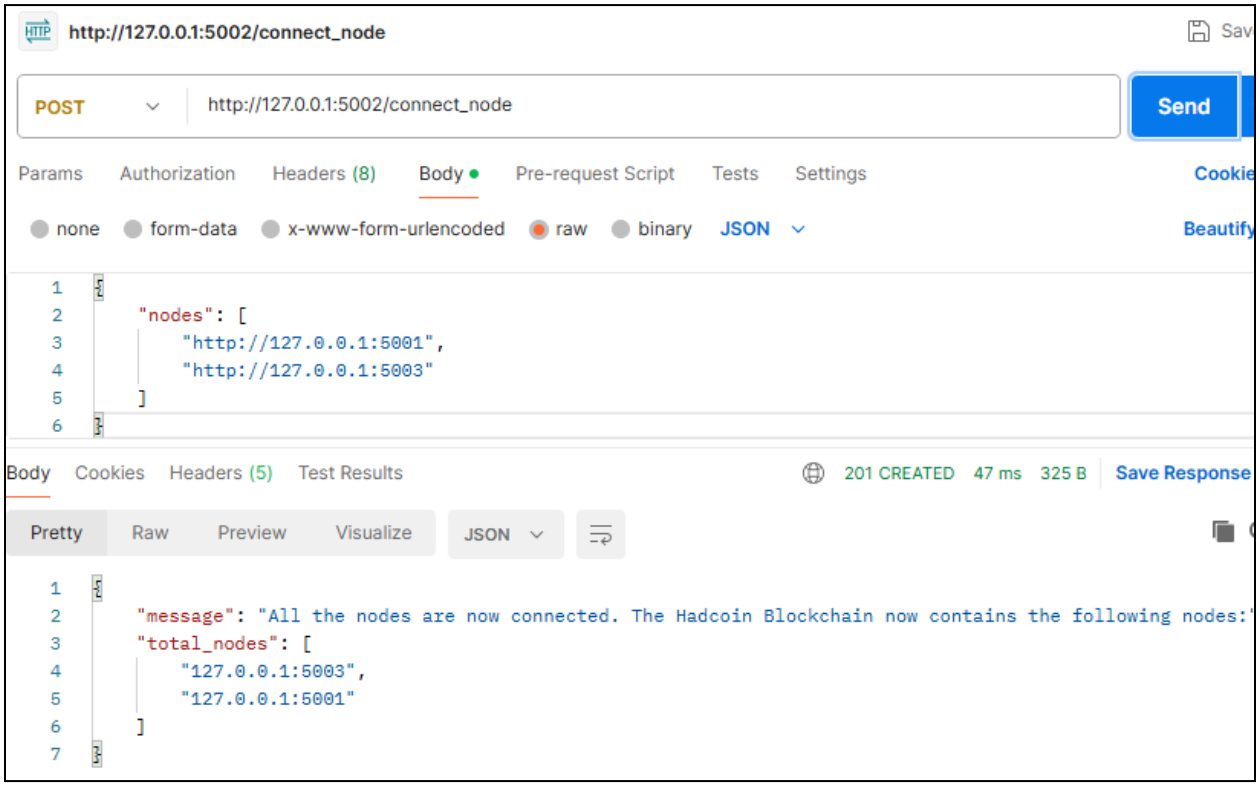
Body Cookies Headers (5) Test Results

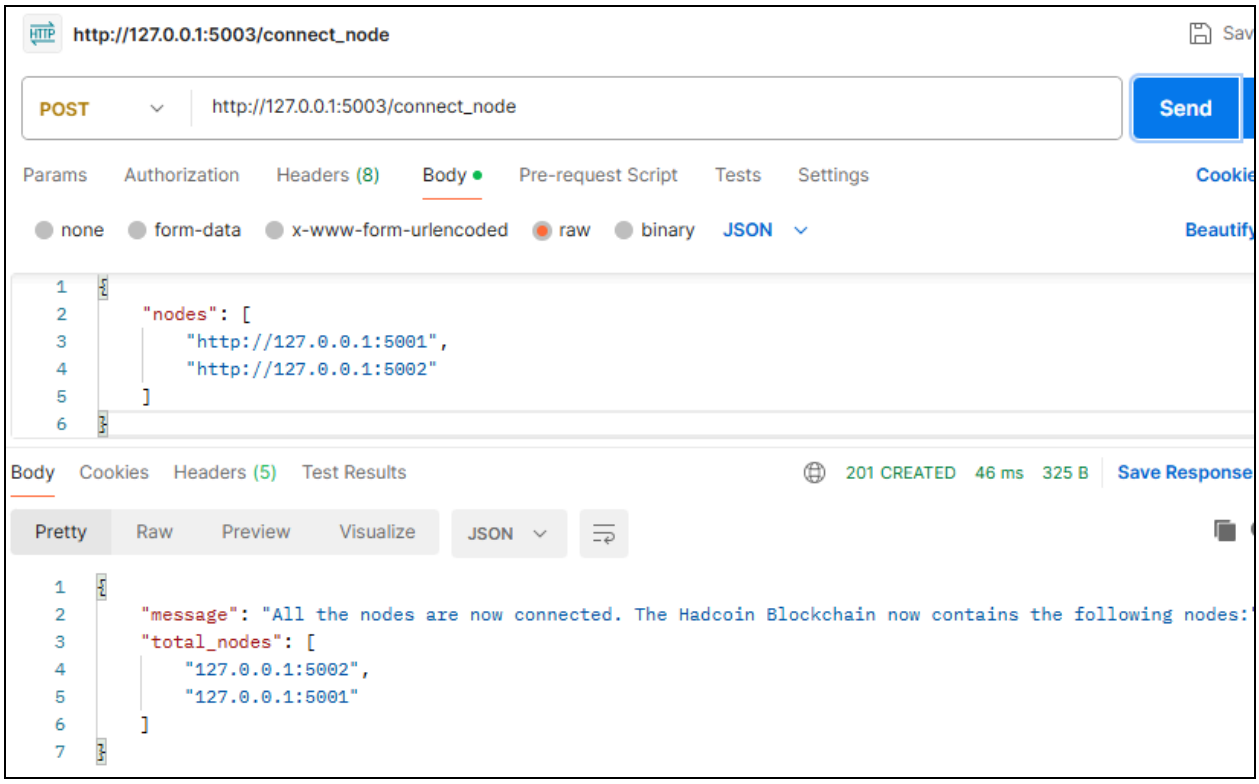
Pretty Raw Preview Visualize JSON


```
{
    "chain": [
        {
            "index": 1,
            "previous_hash": "0",
            "proof": 1,
            "timestamp": "2026-08-24 09:54:23.417238",
            "transactions": []
        }
    ],
    "length": 1
}
```

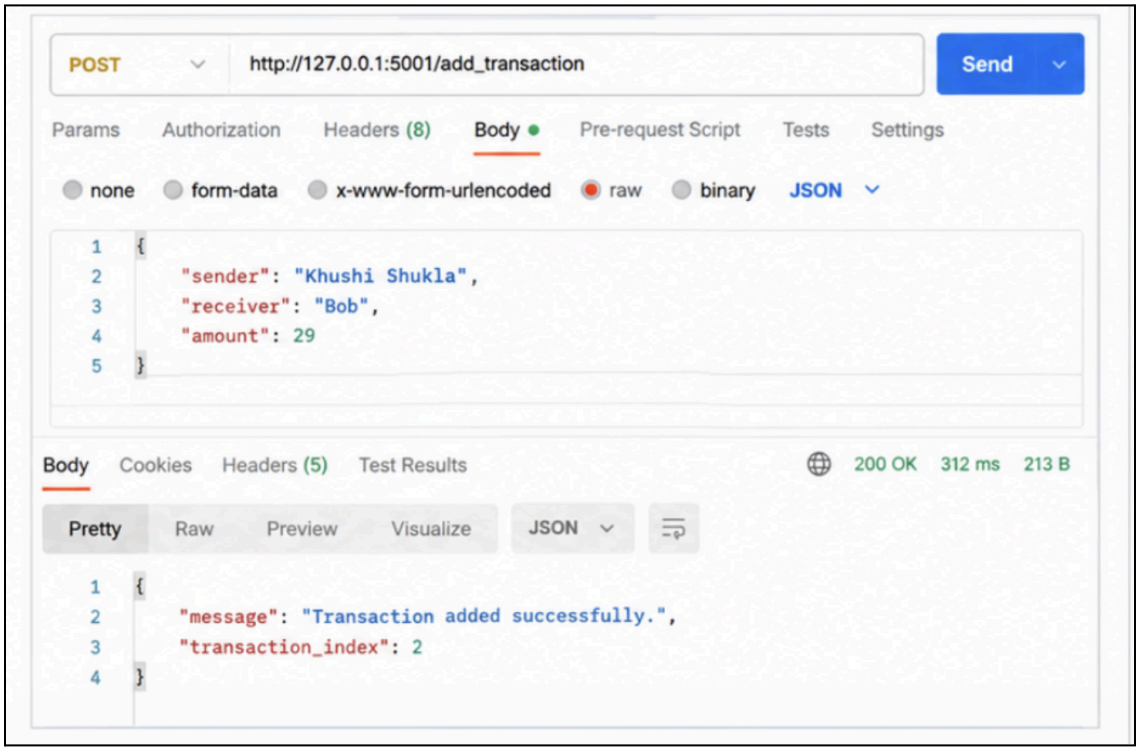
Step 10: Connect the Nodes



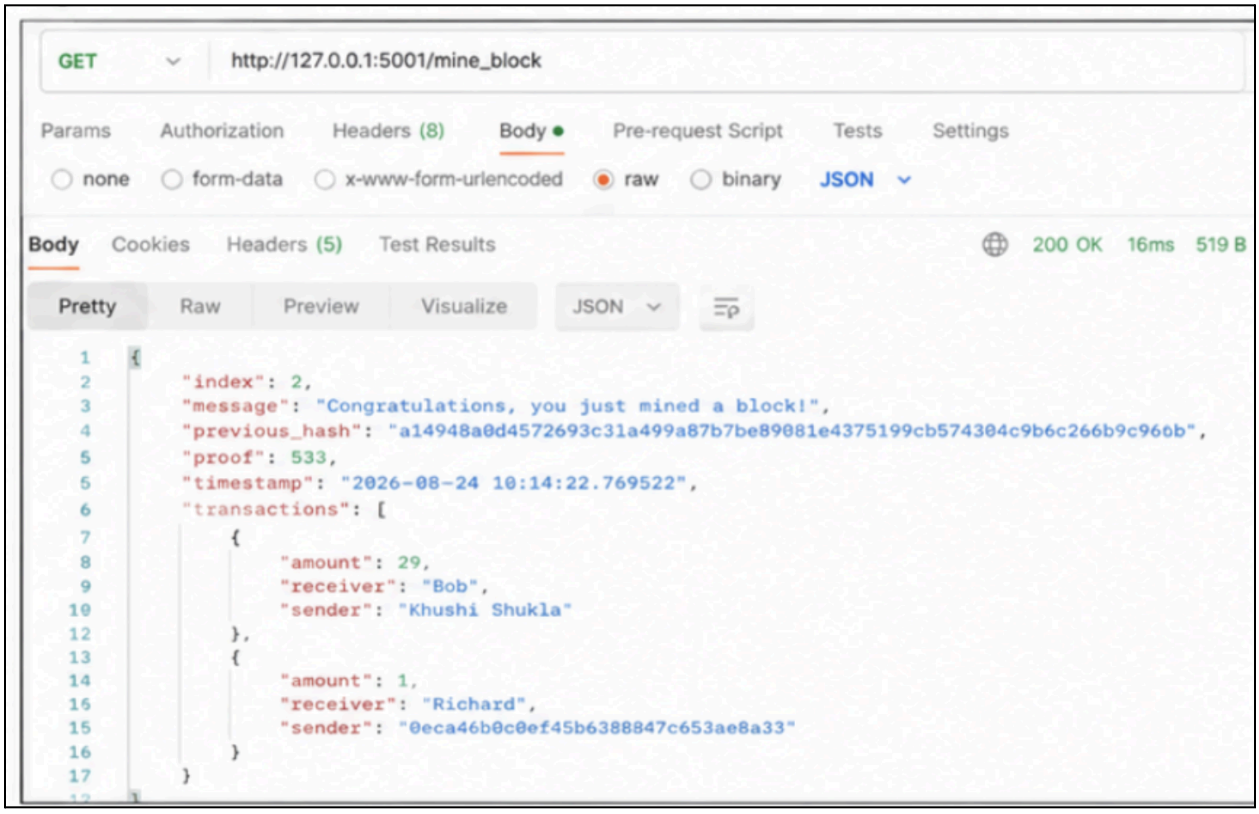




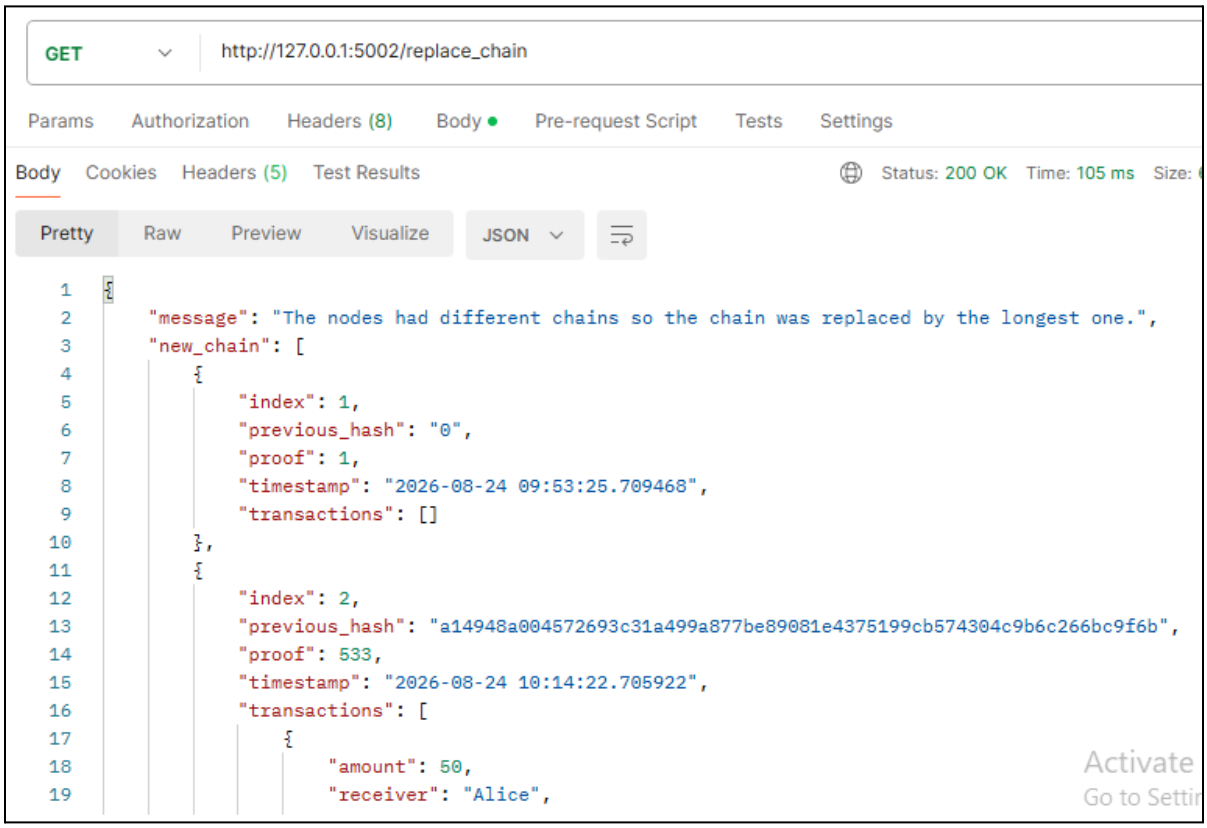
Step 11: Add a Transaction



Step 12: Mine a Block

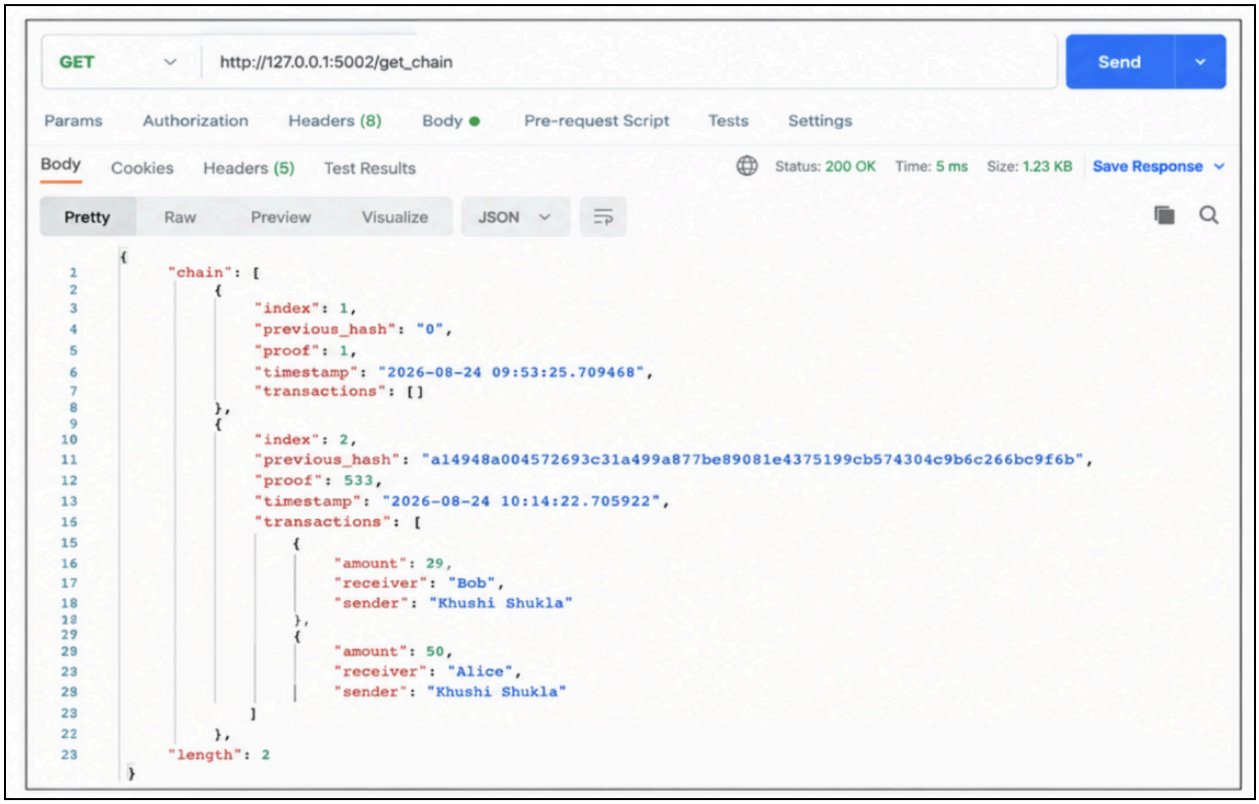


Step 13: Replace the chain using replace_chain()



```
    {
        "amount": 29,
        "receiver": "Bob",
        "sender": "Khushi Shukla"
    },
    {
        "amount": 1,
        "receiver": "Richard",
        "sender": "0eca46b0c0ef45b6388847c653ae8a33"
    }
}
```


Step 14: Fetch the updated chain using get_chain()



Step 12: Verify blockchain using is_valid()

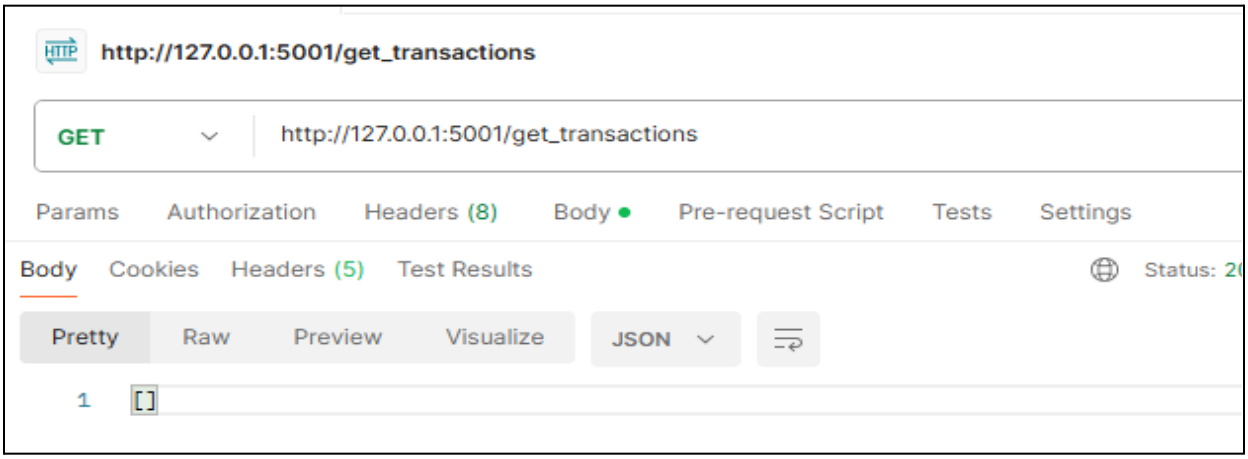
Step 13: Verify that pending transactions are removed after mining

The create_block() function was modified/verified to clear the pending transaction list after the transactions are added to a block.
The following statement was added after assigning transactions to the block:
self.transactions = []

The relevant code is:

```
def create_block(self, proof, previous_hash):
    block = {
        'index': len(self.chain) + 1,
        'timestamp': str(datetime.datetime.now()),
        'proof': proof,
        'previous_hash': previous_hash,
        'transactions': self.transactions
    }
    self.transactions = []
    self.chain.append(block)
    return block
```

This ensures that once pending transactions are included in a block, they are removed from the pending transaction list and are not added again to subsequent blocks.



Conclusion

We developed a cryptocurrency using Python and implemented mining in a simulated three-node blockchain network. Each node maintained its own ledger and synchronized with peers using the Longest Chain Rule to ensure consistency. Mining was performed through Proof-of-Work, securely adding transactions and rewarding miners. This demonstrated key blockchain concepts, including decentralized consensus, transaction validation, and network synchronization.